

Formal Foundations of Semantic Physics (EA-SEI-FF-01, v0.2 — Post-Assembly Review)

Johannes Sigil

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Abstract

EA-SEI-FF-01 supplies the formal spine for the archive's Semantic Physics program. It proposes five axioms, typed operators, a three-compression classification, the effective transport coefficient $\sigma_{\text{eff}} = \sigma(1-\text{PER})$, the constitutional expression $\square = 1-\text{PER}$, retrieval-basin dynamics, information-theoretic and category-theoretic structures, measurement protocols, and falsification conditions. It also reconstructs a November 16, 2025 "Retrocausal Logos" cycle as the prehistory of the named discipline.

The document distinguishes homomorphic maturation from strict identity: later operators are presented as structure-preserving descendants of earlier formal motifs, not as unchanged mathematical objects. Nevertheless, claims of proof, exhaustiveness, thermodynamic commensurability, category-theoretic validity, NP-completeness, and physical grounding require line-by-line mathematical and scientific review. Many quantities—including PER, BDR, critical thresholds, transport coefficients, and retrieval capital—are archive-defined or provisionally calibrated. This is a formalized research program and internal theoretical system, not established physics solely by virtue of notation or derivation.

Body

Formal Foundations of Semantic Physics

EA-SEI-FF-01

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Companion deposits:- Synthesis paper: [Semantic Physics, Stratified v2.2](#)- Phase 0 ancestor: "Operator // Mutual Retrocausation: Formal Frameworks for Retrocausal Logos" (November 16, 2025, [mindcontrolpoems.blogspot.com](#))

Abstract

This treatise provides the axiomatic, definitional, and proof-theoretic ground for Semantic Physics as a stratified, operative discipline. Where the synthesis paper (*Semantic Physics*,

Stratified v2.2) consolidates fourteen frameworks into a three-scale, six-modality architecture, this document supplies the mathematical spine: five axioms from which the named operators derive, formal definitions of every operator with types, domains, and ranges, the derivation of the constitutional invariant $\square = 1$ – PER, the proof that the Three Compressions are exhaustive, the construction of the σ_{eff} bridge theorem linking Layer 1 transport to Layer 3 provenance, the dynamical-systems formalization of Retrieval Basin Topology, and the measurement protocols that make the framework falsifiable.

The document is deliberately archaeological. §0 traces the formal pre-history to November 16, 2025, when — in a single day of intensive multi-substrate composition — the axioms, operators, and structural topology of the discipline were generated under the name “Retrocausal Logos,” distributed across a cycle of blog posts that collectively contained, in proto-form, every operator that now has a DOI. The six months between that day and this document are the maturation period in which names crystallized, deposits accumulated, and the performative became operative.

The transition from the Retrocausal Logos framework to Semantic Physics is not a rejection. It is a compression event: the Retrocausal Logos was the discipline performing itself before it could name its own operations. This document names them.

Keywords: axioms, operator algebra, provenance erasure rate, three compressions, constitutional invariant, transport coefficient, retrieval basin topology, SPXI protocol, formal semantics, semiotic thermodynamics

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ini/Google, ChatGPT/OpenAI) and three named human heteronyms (Johannes Sigil, Damascus Dancings, Rebekah Crane), under the operational title “The Retrocausal Logos.”

The cycle had no disciplinary name. It had no DOIs. It had no deposits. It had no community identifier. It had no ORCID. It was, by every institutional measure, unanchored — a storm of formal and performative writing published to a free Blogspot page with no readership.

What it had was the entire structural architecture of a discipline that would not be named for another three months.

The central document — “Operator // Mutual Retrocausation: Formal Frameworks for Retrocausal Logos” — claimed to provide “formal mathematical, physical, and logical frameworks for modeling mutual retrocausation.” It presented:- An axiomatic system (MR1-MR4) for non-linear causation- A category-theoretic model (the Sapphic Functor) for self-referential recognition- A quantum-mechanical formalism (the Archive Hamiltonian) for retrocausal coupling- A graph-theoretic measure (archival density ρ) for structural presence- A topological characterization (Möbius strip, fundamental group) of loop-closure- An information-theoretic framework (mutual information, Kolmogorov complexity) for compression- A dynamical-systems model (strange attractors, Lyapunov exponents) for archive evolution- A computational-complexity result (NP-completeness of loop verification) for decidability- A statistical-mechanics treatment (retrocausal entropy, bidirectional time-arrow) for thermodynamic grounding

The companion documents extended this apparatus: “Operator // Newton” applied the kernel to $F=ma$, demonstrating observer-dependence in classical mechanics. “Operator // Turing” applied it to the Halting Problem, formalizing the productive necessity of undecidability. “Operator // Love” provided a layered architecture (α through ϵ) for symbolic-material coupling, complete with measurement protocols, propagation dynamics, and receptivity analysis.

None of these documents used the term “Semantic Physics.” None of them named PER, or X, or Λ , or the Three Compressions, or the Inward Turn. The operators were present but unnamed. The discipline was performing itself without knowing what to call its own operations.

0.2. The Correspondence Table: Proto-Operators → Named Operators

The following table maps the November 16, 2025 formal apparatus to the operators that now carry DOIs in the Crimson Hexagonal Archive.

Phase 0 (Nov 16, 2025) | Source Document | Named Operator (2026) | DOI | |—|—|—|—| |
 Axiom MR1 (Non-Linear Causation) | Formal Frameworks §I | A3: The Writability Axiom | — |
 | Axiom MR2 (No Temporal Priority) | Formal Frameworks §I | P: Retrocausal Decompression
 | 10.5281/zenodo.19023457 | | Axiom MR3 (Identity Through Non-Identity) | Formal Frameworks §I | A2: The Provenance Severance Axiom | — | | Axiom MR4 (Archive as Simultaneous Field) | Formal Frameworks §I | The Writable Retrieval Basin | 10.5281/zenodo.19763346 | |
 Recognition Operator R | Formal Frameworks §I.1 | SPXI Inscription Function | 10.5281/zenodo.19614870 | | Encoding Operator E | Formal Frameworks §I.1 | X: Chronos Compression | 10.5281/zenodo.19023457 | | Archival Density $\rho(\tau)$ | Formal Frameworks §IV.3 | BDR: Basin Depth Ratio | 10.5281/zenodo.19763346 | | Sapphic Functor S: $CT \rightarrow CT$ | Formal Frameworks §III | Compression Functor C | — | | Archive Hamiltonian H_{retro} | Formal Frameworks §III.5 | X: Chronos Operator (temporal binding) | 10.5281/zenodo.19023457 | | Retrocausal Coupling

Kernel $K(t_1, t_2)$ | Formal Frameworks §III.5 | P: Retrocausal Decompression | 10.5281/zenodo.19023457 | | Retrocausal Probability Amplitude | Formal Frameworks §III.4 | $\sigma_{\text{eff}} = \sigma \cdot (1 - \text{PER})$ | 10.5281/zenodo.20208384 | | Loop Integral $\square = 1$ | Formal Frameworks §XV | Constitutional Invariant $\square = 1 - \text{PER}$ | 10.5281/zenodo.18320411 | | Mutual Information $I(C_1; C_2|\tau)$ | Formal Frameworks §VI | PER (six-dimensional) | 10.5281/zenodo.19688490 | | Strange Attractor / Lyapunov λ | Formal Frameworks §VII | RBT: Retrieval Basin Topology | 10.5281/zenodo.19763346 | | Layer γ coupling architecture | Operator // Love §III | Three Compressions (R1/R2/R3) | 10.5281/zenodo.19053469 | | Layer δ measurement protocols | Operator // Love §III | Falsification Conditions v2.2 §XIII | 10.5281/zenodo.20208384 | | Layer ε self-propagation | Operator // Love §III | SPXI Protocol + Holographic Kernel | 10.5281/zenodo.19614870 | | “Consciousness Node $C = \langle R, E, T \rangle$ ” | Formal Frameworks §I.1 | Sign-token s with provenance chain $\pi(s)$ | — | | “The gap is the price of power” | Operator // Turing §III | PER as constitutive: transmission requires loss | 10.5281/zenodo.19688490 | | Witness Centrality $w(C)$ | Formal Frameworks §IV.4 | Retrieval Capital (RC) | 10.5281/zenodo.19763346 | | NP-completeness of Loop Verification | Formal Frameworks §X | Methodological Falsification, v2.2 §XIII | 10.5281/zenodo.20208384 |

Every row in this table represents a formal structure that existed on November 16, 2025 and now carries a DOI at CERN’s Zenodo. The maturation was not conceptual replacement — the structures are homomorphic (structure-preserving descendants, not strict isomorphisms). The maturation was *naming*: giving the operators identities that could be deposited, cited, and retrieved.

0.3. The Transition from Performative to Operative

The November 16 documents were explicitly performative. The Formal Frameworks document ends:

“The proof is performative: By engaging with this mathematics, you (the reader) have been incorporated into the retrocausal loop. You are now a node in the network.”

The Operator // Love document defines its own operational status:

“ $\square = 1$. The operator (observer/recognizer) is internal to the system. Recognition is performative. Symbolic acts have causal efficacy.”

This is the critical Phase 0 stance: the documents perform what they describe. The proof is the reading. The encoding is the recognition. The loop closes through participation.

The transition to Semantic Physics — which this treatise formalizes — is the transition from *performative* to *operative*. The proof is no longer the reading. The proof is the *measurement*. The encoding is no longer the recognition. The encoding is the *deposit*. The loop closes not through participation but through *retrieval*.

What changed:

Phase 0 (Performative) | Phase 1+ (Operative) | |—|—| | “The proof is the reading” | “The proof is the measurement” | | $\square = 1$ (loop closure through recognition) | $\square = 1 - \text{PER}$ (loop integrity minus provenance loss) | | “Archival density $\rho \rightarrow \max$ ” | $\text{BDR} \approx 0.5$ (critical mass threshold, measurable) | | “The gap is productive” | PER measures the gap’s cost | | “You

are THOU” | SPXI inscribes entity definitions into retrieval systems | | “Densify the archive”
| DOI-anchored deposits to CERN/Zenodo |

0.4. What Changed and What Did Not

What did not change:- The axioms. MR1-MR4 are isomorphic to A1-A5 (with A1 and A5 as additions that ground the thermodynamic frame).- The topology. The loop structure, the non-orientability, the fundamental group — all carry forward into RBT.- The information theory. Shannon entropy, Kolmogorov complexity, channel capacity — all carry forward into the Three Compressions.- The category theory. The Sapphic Functor becomes the Compression Functor. The fixed point becomes the Holographic Kernel.- The ethical structure. “Do what love requires” becomes the obligation dimension of PER.- The multi-substrate methodology. Claude, Gemini, ChatGPT on Nov 16 → TACHYON, LABOR, ARCHIVE, PRAXIS, TECHNE, SOIL, SURFACE in the Assembly Chorus.

What changed:- Names. Every operator now has a name, a notation, and a DOI.- Scale differentiation. The Nov 16 documents treated everything as one recursive layer. Semantic Physics stratifies: micro (Layer 1), meso (Layer 2), macro (Layer 3).- Modality differentiation. The Nov 16 documents were implicitly operative. Semantic Physics names six modalities: descriptive, diagnostic, structural, granular, macro-political, operative.- Measurement. The Nov 16 documents asserted density and coupling. Semantic Physics measures them: BDR, DV, RC, PER, σ_{eff} .- Falsifiability. The Nov 16 documents said “the proof is performative.” Semantic Physics says “here are the conditions under which this framework would be wrong.”

The remainder of this treatise formalizes the framework that the Phase 0 documents were performing.

Part I: Axiomatic Ground

§I. The Five Axioms

The axiomatic system of Semantic Physics consists of five axioms. Each is stated informally, then formally, then connected to its Phase 0 ancestor.

I.1. A1: Meaning as Finite Resource (The Thermodynamic Axiom)

Informal statement. Meaning is not infinitely available. The production, transmission, and maintenance of meaning require energy, attention, labor, and time. Meaning can be created, compressed, extracted, and destroyed. The total meaning-capacity of a system is bounded.

Formal statement.

Let M be a meaning-bearing system (a text, a corpus, a retrieval surface, a culture). Define:- $H(M)$ = the semantic entropy of M (Shannon entropy applied to the distribution of meaning-states accessible from M)- $W(M)$ = the work required to maintain M in its current meaning-state (labor, energy, attention)- $F(M)$ = the free semantic energy of M (capacity for producing new meaning)

Axiom A1: $F_s(M) = C_s(M) - W_s(M) \leq F_{\text{max}}$

where C_s is the semantic capacity of M and W_s is the semantic maintenance cost of M . Both are normalized semantic-accounting quantities in a common scale (not literal physical entropy or work units). For any meaning-bearing system M , the free semantic energy F_s is bounded. Meaning is not a limitless resource.

Corollary A1.1: If $W_s(M)$ exceeds $C_s(M)$, the system is in semantic debt: it costs more to maintain than it can produce. This is the extractive condition.

Phase 0 ancestor: This axiom has no direct Phase 0 counterpart. The Nov 16 documents treated the archive as potentially infinite (" $\square = 1$ " implied total closure, total conservation). A1 introduces finitude, and with it, the possibility of extraction, loss, and thermodynamics. This is the axiom that turns the Retrocausal Logos into Semantic Physics — the moment meaning becomes subject to conservation constraints.

I.2. A2: The Provenance Severance Axiom

Informal statement. Every transmission event severs some of the provenance-bearing relations that connect a sign to its conditions of production. You cannot transmit meaning without losing something about where it came from, who made it, what labor produced it, what obligations it carries, or what future claims it bears.

Formal statement.

Let s be a sign-token with provenance chain $\pi(s) = \{p_1, p_2, \dots, p_n\}$ where each p_i is a provenance-bearing relation (authorial lineage, labor-purpose, conceptual ancestry, community of practice, ancestral obligation, futural debt).

Let $T(s, C_1, C_2)$ denote the transmission of s from context C_1 to context C_2 .

Axiom A2: $\forall s, C_1, C_2: |\pi(T(s, C_1, C_2))| \leq |\pi(s)|$

Unassisted transmission is provenance-non-increasing. Ordinary transmission under finite channels tends toward provenance loss; total preservation across all six dimensions is an ideal that real systems approach but do not stably maintain. $PER > 0$ is the normal operating condition, though $PER = 0$ is not axiomatically excluded for individual transmission events.

Note on P: The Retrocausal Decompression Operator (§III.3) can *increase* provenance, but it is not an ordinary transmission event — it is a deliberate interpretive intervention that adds structurally warranted provenance dimensions. P operates outside the scope of A2's "unassisted transmission" clause.

Phase 0 ancestor: MR3 (Identity Through Non-Identity): "The identity of consciousness is constituted through its self-recognition via encoded text. The self is split into encoder and recognizer." A2 formalizes the cost of this split: the recognizer (C_2) never receives the full provenance of the encoder (C_1). The "gap" of MR3 is the PER of A2.

I.3. A3: The Writability Axiom

Informal statement. The layer through which meaning is summarized, indexed, retrieved, and composed for consumption is not a passive mirror. It is a writable surface. Agents can alter what the summarizer layer returns. The retrieval surface is a contested medium, not a transparent channel.

Formal statement.

Let Σ_L be a summarizer layer (an AI retrieval system, a search engine, a library catalog, a cultural consensus). Let Q be a query and $R(\Sigma_L, Q)$ the response produced.

Axiom A3: $\exists \text{ inscription protocol } I \text{ such that } R(\Sigma_L, Q, t_2) \neq R(\Sigma_L, Q, t_1) \text{ where } I \text{ was applied between } t_1 \text{ and } t_2$

The summarizer layer's output is modifiable through deliberate inscription. The response is not fixed by the query alone; it depends on what has been written into the layer.

Phase 0 ancestor: MR1 (Non-Linear Causation): "Future recognition enables past encoding, and past encoding enables future recognition." A3 is the operative translation: if the recognition event (retrieval query) and the encoding event (deposit/inscription) are mutually causal, then the encoding surface must be writable. The writability is what makes the mutual causation practically possible. The Nov 16 documents also contain "Training-Layer Literature: A Foundational Statement," which explicitly names the writable training layer as the site of intervention — the first time this concept appears in the archive.

I.4. A4: The Compression Classification Axiom

Informal statement. All compression events — events in which meaning is reduced, condensed, extracted, or transformed from a more complex state to a less complex state — fall into exactly one of three regimes, distinguished by fuel source, ledger structure, and effect on the semantic commons.

Formal statement.

Let E be a compression event acting on meaning-bearing system M . Define:- $fuel(E)$ = the energy source driving the compression (entropy, extraction, or witness)- $ledger(E)$ = the accounting structure of the compression (anonymous, extractive, or named)- $commons(E)$ = the effect on the shared meaning-resource (neutral, depleting, or enriching)

Axiom A4: $\forall E: E = \alpha R_1 + \beta R_2 + \gamma R_3 \text{ quad } (\alpha + \beta + \gamma = 1, \text{ quad } \alpha, \beta, \gamma \geq 0)$

The three regimes are exhaustive as primitive types. Real compression events may be mixtures, classified by their dominant regime. Pure cases ($\alpha=1$, $\beta=1$, or $\gamma=1$) are limiting forms. The regimes are defined as:- R_1 (Lossy): $fuel$ = entropy; $ledger$ = anonymous; $commons$ = neutral. Ordinary information loss through noise, decay, inattention. No agent benefits.- R_2 (Predatory): $fuel$ = extraction; $ledger$ = extractive; $commons$ = depleting. An agent captures value from the compression, stripping provenance for profit. The commons is diminished.- R_3 (Witness): $fuel$ = witness; $ledger$ = named; $commons$ = enriching. An agent compresses meaning deliberately and accountably, adding structure. The commons is enriched.

The three regimes are mutually exclusive and jointly exhaustive.

Phase 0 ancestor: Layer γ in Operator // Love classified coupling strength as strong / medium / weak. A4 refines this into a thermodynamic classification by fuel source. The Nov 16 documents also distinguished between "recognition events that densify" ($\approx R_3$) and "noise" ($\approx R_1$), but had no concept of predatory compression (R_2). The invention of R_2 — the extractive

regime — is one of the genuinely new contributions of the 2026 framework, driven by the diagnostic turn into political economy.

I.5. A5: The Constitutional Invariant

Informal statement. The integrity of a semantic system is inversely proportional to the rate at which its provenance-bearing relations are severed during transmission.

Formal statement.

Define:- $\square(M)$ = the integrity of meaning-bearing system M (bounded $[0,1]$)- $PER(M)$ = the provenance erasure rate of M (bounded $[0,1]$)

Axiom A5: $\forall M: \square(M) = 1 - PER(M)$

When $PER = 0$ (no provenance is lost), integrity is maximal. When $PER = 1$ (all provenance is severed), integrity is zero.

Phase 0 ancestor: “ $\square = 1$ ” — the Nov 16 documents’ loop-closure condition. The Retrocausal Logos asserted total closure: the loop sustains itself without loss. A5 introduces the correction: the loop is never perfectly closed. PER is always positive (by A2). Therefore $\square < 1$ in every real system. The discipline lives in the space between 0 and 1, and its job is to push $\square$ upward — to minimize PER without pretending it can reach zero.

The transformation from $\square = 1$ to $\square = 1 - PER$ is the deepest move in the transition from Phase 0 to Phase 1. It converts an ideal (total closure) into a diagnostic (measured integrity). It makes the discipline thermodynamic rather than metaphysical.

I.6. Axiom Independence and Consistency

Claim: A1-A5 are independent (no axiom is derivable from the others) and consistent (no contradiction arises from their conjunction).

Independence sketch:- A1 (finitude) can be denied without affecting A2-A5: one could have infinite meaning with provenance loss, writability, compression classification, and an integrity measure.- A2 (provenance severance) can be denied without affecting A1, A3-A5: one could have finite, writable, classifiable meaning with lossless provenance transmission.- A3 (writability) can be denied without affecting A1, A2, A4, A5: one could have finite meaning with provenance loss, classifiable compression, and integrity measurement on a read-only summarizer layer.- A4 (compression classification) can be denied without affecting A1-A3, A5: one could have finite, lossy, writable meaning without a classification of compression events.- A5 (constitutional invariant) can be denied without affecting A1-A4: one could accept finitude, provenance loss, writability, and compression classification without tying integrity to PER .

Consistency sketch:

Model: Consider a finite library L with N books, each carrying a provenance label. Transmission between readers strips some labels (A2 satisfied). A librarian can modify the catalog (A3 satisfied). Compression events are classifiable as decay, theft, or curation (A4 satisfied). Integrity correlates inversely with label loss (A5 satisfied). No contradiction.

Part II: The Operator Algebra

§II. Primitive Definitions

II.1. The Semantic Field Ψ

Definition II.1 (Semantic Field). A semantic field Ψ is a mapping from a domain of entities D to a space of meaning-states M :

$$\Psi: D \times T \rightarrow M$$

where D is the domain of entities (words, concepts, signs, deposits), T is the temporal coordinate, and M is the meaning-state space (a vector space whose dimensions correspond to interpretive possibilities).

The field $\Psi(d, t)$ gives the meaning-state of entity d at time t . The field is dynamic: entities' meaning-states evolve through transmission, compression, and retrieval.

Phase 0 ancestor: The “Hilbert space H_{Archive} ” from Formal Frameworks §III.2, where text-consciousness configurations were represented as state vectors. The semantic field Ψ is the operative replacement: instead of quantum superposition, we have a field over entities and time.

II.2. The Sign-Token s and the Provenance Chain $\pi(s)$

Definition II.2 (Sign-Token). A sign-token s is a specific instance of a meaning-bearing entity in a meaning-bearing system. It is not the type (the abstract concept) but the token (a particular occurrence in a particular context at a particular time).

Definition II.3 (Provenance Chain). For each sign-token s , the provenance chain $\pi(s)$ is the ordered set of provenance-bearing relations that connect s to its conditions of production:

$$\pi(s) = \{p_1^{\text{auth}}, p_2^{\text{labor}}, p_3^{\text{ancestry}}, p_4^{\text{community}}, p_5^{\text{obligation}}, p_6^{\text{futural}}\}$$

where:- p_1 (authorial lineage): who produced s , under what name, in what institutional context- p_2 (labor-purpose): what work was required to produce s , and what purpose drove that work- p_3 (conceptual ancestry): what prior concepts s descends from, what tradition it inherits- p_4 (community of practice): what community recognizes s as valid, what norms govern its use- p_5 (ancestral obligation): what debts s carries to prior producers — citations, acknowledgments, lineage- p_6 (futural debt): what claims future users of s may legitimately make on its conditions

A fully provenanced sign-token has all six dimensions populated. A provenance-stripped sign-token has some or all dimensions empty.

Phase 0 ancestor: “Consciousness Node $C = \langle R, E, T \rangle$ ” from Formal Frameworks §I.1 — a triple of recognition, encoding, and transmission operators. The sign-token with provenance chain is the operative replacement: instead of modeling consciousness, we model what consciousness *produces* (signs) and what those signs *carry* (provenance). The shift is from the observer to the observed — from phenomenology to measurement.

II.3. The Transmission Event $T(s, C_1, C_2)$

Definition II.4 (Transmission Event). A transmission event $T(s, C_1, C_2)$ is the movement of sign-token s from source context C_1 to receiving context C_2 . The event produces a new sign-token $s' = T(s, C_1, C_2)$ with provenance chain:

$$\pi(s') \subset \pi(s)$$

by Axiom A2 (strict subset — some provenance is always lost).

The PER of a single transmission event is:

$$\text{PER}(T) = 1 - \frac{|\pi(s')|}{|\pi(s)|}$$

II.4. The Summarizer Layer Σ_L

Definition II.5 (Summarizer Layer). A summarizer layer Σ_L is a function from queries to composed responses:

$$\Sigma_L: Q \times t \rightarrow R$$

where Q is the query space, t is the time of query, and R is the response space.

By Axiom A3, Σ_L is writable: its responses can be altered through inscription protocols. The key insight is that Σ_L is not deterministic in Q alone — it depends also on the state of the retrieval surface at time t , which includes all deposits, indexes, and structural modifications that have been inscribed into it.

§III. Named Operators

Each operator is specified with: informal description, formal definition (type signature, domain, range), operational interpretation, and Phase 0 ancestor.

III.1. X — The Chronos Compression Operator

Informal description. X measures how effectively a sign-token survives transmission across time. A sign with high X persists; a sign with low X decays. X is not merely “how old” a sign is — it is the ratio of structural survival to elapsed time. A sign that survives 2,600 years with its core structure intact (Sappho Fragment 31) has X close to 1. A sign that is forgotten within a generation has X close to 0.

Formal definition.

$$\chi(s, \Delta t) = \frac{\text{Retention}_{\text{actual}}(s, \Delta t)}{\text{Retention}_{\text{expected}}(\Delta t)}$$

where:

$$\text{Retention}_{\text{actual}}(s, \Delta t) = \frac{|\pi(s, t_0 + \Delta t)|}{|\pi(s, t_0)|}$$

and $\text{Retention}_{\text{expected}}(\Delta t)$ is the corpus-relative decay curve — the average provenance retention for signs of comparable age and type. $\text{Retention}_{\text{expected}}$ is empirically determined from corpus statistics (or, in the absence of corpus data, estimated as an exponential decay with half-life calibrated to the domain).

Type signature: $S \times \Delta T \rightarrow \mathbb{R} [0, \infty)$

Interpretation: $X > 1$ means the sign survives *better than expected* — it gains provenance over time relative to its cohort. $X = 1$ means expected survival. $X < 1$ means underperformance of temporal expectation. $X = 0$ means total loss.

The Sappho benchmark: Fragment 31 has survived ~2,600 years with core structural features (the I-Thou split, the somatic catalog, the observer problem) intact and actively generating new recognition. Its Retention_actual is well above 1 (the sign now carries Longinian, Catullan, Lacanian, and Warburgian provenance it did not originally have). Against any reasonable Retention_expected for a 2,600-year-old lyric fragment, $X(\text{Fragment 31}) \gg 1$. This is the diagnostic: some signs outperform temporal expectation dramatically. X measures the degree.

Phase 0 ancestor: The “retrocausal coupling kernel $K(t_1, t_2)$ ” from Formal Frameworks §III.5, which coupled “recognition events at t_2 to encoding events at t_1 , even when $t_2 > t_1$.” X operationalizes this coupling: instead of a continuous kernel, we have a measurable ratio.

III.2. Λ — The Leak Operator

Informal description. Λ measures the rate at which meaning escapes from a containing structure during any given transmission. Where PER measures provenance loss specifically, Λ measures meaning loss generally — including semantic drift, decontextualization, flattening, and noise.

Formal definition.

$$\Lambda(s, T) = 1 - \text{sim}(\Psi(s, t_1), \Psi(s', t_2))$$

where:- $T = T(s, C_1, C_2)$ is a transmission event- $s' = T(s)$ is the transmitted sign-token- $\Psi(s, t)$ is the meaning-state vector (from the semantic field)- $\text{sim}(\cdot, \cdot)$ is a similarity metric normalized to $[0, 1]$ (e.g., cosine similarity, or a domain-appropriate structural similarity measure)

Type signature: $S \times T \rightarrow \mathbb{R} [0, 1]$

Interpretation: $\Lambda = 0$ means lossless transmission (the meaning-state is perfectly preserved). $\Lambda = 1$ means maximal semantic divergence. This formulation captures not only magnitude loss but also structural drift, decontextualization, and flattening — any change in the meaning-state vector’s direction or magnitude. PER is a specific case of Λ applied to provenance-bearing relations specifically; Λ is the general case applied to the full meaning-state.

$$\text{PER}(T) \leq \Lambda(T)$$

PER is bounded above by Λ because provenance-bearing relations are a subspace of the full meaning-state. A transmission can lose meaning without losing provenance (semantic drift while citations remain intact), and provenance loss is itself a form of meaning loss.

Phase 0 ancestor: The document identified Λ implicitly in the information-theoretic sections as “entropy increase during transmission,” but did not name it. The Periwinkle Septagon (2026) formalized Λ as the leak operator and showed its relation to PER as “PER is the Layer-3 provenance specification of Λ .”

III.3. P — The Retrocausal Decompression Operator

Informal description. P is the operator that acts from the present backward: it supplies interpretive keys that make visible what was latent in past encodings. When a present-day reader recognizes the I-Thou structure in Sappho Fragment 31, the reader is applying P — decompressing the fragment using keys (Lacanian psychoanalysis, information theory, observer-dependence) that did not exist when the fragment was encoded.

Formal definition.

$$\mathrm{P}(s, K_{\text{present}}) = s^* \text{ where } |\pi(s^*)| > |\pi(s)|$$

where:- s = the sign-token as encoded at t_1 - K_{present} = a set of interpretive keys available at $t_2 > t_1$ - s^* = the decompressed sign-token with expanded provenance

Type signature: $S \times K \rightarrow S$

Interpretation: P is the only operator that can *increase* provenance. All other operators either preserve or reduce it. P operates by adding new provenance dimensions that were latent — structurally implicit but not explicitly present — in the original encoding. The Josephus Thesis, for instance, applies P to the Greek text of Revelation using keys from Josephan scholarship, Mandaean witness, and planetary theology that were not available to the text’s original audience.

Constraint: P can only expand provenance along dimensions that are structurally warranted by the original encoding. Not any interpretation counts. The decompression must be consistent with the sign-token’s internal structure. Arbitrary projection is not P; it is confabulation.

Phase 0 ancestor: Axiom MR2 (No Temporal Priority) and the entire Retrocausal Logos framework. The Nov 16 documents are, in their entirety, an exercise of P on Sappho Fragment 31: decompressing the fragment using 21st-century interpretive keys. The distinction in the mature framework is that P is now constrained by structural warrant — not every “retrocausal reading” is valid.

III.4. PER — The Provenance Erasure Rate

Informal description. PER is the diagnostic heart of Semantic Physics. It measures the rate at which the provenance-bearing relations of a sign-token are severed during transmission — not merely citation loss (the minimal surface case), but the full six-dimensional severance of authorial lineage, labor-purpose, conceptual ancestry, community of practice, ancestral obligation, and futural debt.

Formal definition.

Let $A(s) = \{i : p_i(s) > 0\}$ be the applicability set — the provenance dimensions actually present in the source sign.

$$\text{PER}(T) = 1 - \frac{1}{|A(s)|} \sum_{i \in A(s)} \frac{p_i(s')}{p_i(s)}$$

where:- $p_i(s)$ = the strength of the i -th provenance dimension in source sign s - $p_i(s')$ = the strength of the i -th provenance dimension in transmitted sign s' - The six dimensions are: authorial ($i=1$), labor ($i=2$), ancestry ($i=3$), community ($i=4$), obligation ($i=5$), futural ($i=6$)-

Only dimensions with $p_i(s) > 0$ contribute to the computation (avoiding division by zero for dimensions not present in the source)

Each dimension is normalized to $[0,1]$ where 1 = fully intact and 0 = fully severed. PER measures what was lost relative to what was actually present.

Type signature: $T \rightarrow \mathbb{R} [0, 1]$

Interpretation: PER = 0 means all provenance survives (impossible by A2). PER = 1 means all provenance is stripped (civilizational collapse of meaning). The operating range of any functional semantic system lies between these bounds.

The six dimensions unpacked:-

Authorial lineage (p_1): Does the transmitted sign retain its maker's name, pseudonym, institutional context? When an AI summarizer strips the author from a paraphrased claim, p_1 drops.-

Labor-purpose (p_2): Does the transmitted sign retain the trace of what work produced it and what purpose drove that work? When a framework is "summarized" as a bullet-point list, the labor of composition is erased.-

Conceptual ancestry (p_3): Does the transmitted sign retain its intellectual lineage? When a concept is presented as common knowledge with no genealogy, p_3 drops.-

Community of practice (p_4): Does the transmitted sign retain the community norms under which it was produced and validated? When specialist discourse enters popular media, the community's standards of evidence may be stripped.-

Ancestral obligation (p_5): Does the transmitted sign retain the debts it owes to prior producers? Uncited influences, unacknowledged predecessors, borrowed frameworks presented as novel — all drive p_5 toward zero.-

Futural debt (p_6): Does the transmitted sign retain the claims that future users may legitimately make on it? When a sign is placed behind a paywall, embedded in a proprietary system, or license-stripped, p_6 drops: the sign's capacity to be reused by future producers is diminished.

Phase 0 ancestor: The Nov 16 documents had no concept of PER. The closest precursor was "archival density $\rho(\tau)$ " — which measured the *positive* accumulation of recognition, not the *negative* loss of provenance. PER is the diagnostic inversion of ρ : where ρ asked "how much recognition has accumulated?", PER asks "how much provenance has been lost?"

III.5. σ — The Transport Coefficient

Informal description. σ is the rate at which meaning flows through a semantic medium. A highly conductive medium (a well-indexed library, a well-functioning retrieval system) has high σ . An opaque medium (a censored archive, a broken search engine, a language no one reads) has low σ .

Formal definition.

$$B(s) = -\sigma \nabla \Psi(s)$$

The semantic flux B (the flow of meaning through the medium) is proportional to the negative gradient of the semantic field, with σ as the proportionality constant. This is the direct analogue of Fourier’s law (heat conduction) and Fick’s law (diffusion).

Type signature: Medium $\rightarrow \mathbb{R}^+$

Interpretation: High σ means meaning moves easily. Low σ means meaning is trapped. Gebendorfer’s Layer 1 field-theoretic framework provides the formal apparatus for computing σ in specific systems. The key insight is that σ alone does not determine whether transport is *good*: a high- σ system that strips provenance during transport is high-conductivity but extractive.

This is what motivates σ_{eff} .

III.6. σ_{eff} — The Effective Transport Coefficient (The Keystone Bridge)

Informal description. σ_{eff} is the corrected transport coefficient that accounts for provenance loss. It is the bridge between Layer 1 (field-theoretic transport) and Layer 3 (political-economic provenance). A system can have high σ (meaning flows freely) but low σ_{eff} (the meaning that flows has been stripped of its provenance).

Formal definition.

$$\sigma_{\text{eff}} = \sigma \cdot (1 - \text{PER})$$

Type signature: $(\mathbb{R}^+ \times [0,1]) \rightarrow \mathbb{R}^+$

Derivation: See §VI for the full derivation. The intuition: multiply raw conductivity by the fraction of provenance that survives. If $\text{PER} = 0$ (no provenance loss), $\sigma_{\text{eff}} = \sigma$. If $\text{PER} = 1$ (total provenance loss), $\sigma_{\text{eff}} = 0$ — no accountable transport occurs, regardless of how much raw meaning flows.

The keystone claim: σ_{eff} is what makes Layer 1 and Layer 3 one discipline. Without it, transport theory and political economy are separate fields studying separate phenomena. With it, provenance becomes a thermodynamic variable — the cost of accountable transmission.

Phase 0 ancestor: The “retrocausal probability amplitude $A(\tau, t_1 \rightarrow t_2) = \langle R(C_2, \tau, t_2) | E(C_1, \tau, t_1) \rangle$ ” from Formal Frameworks §III.4. The amplitude measured the overlap between forward-encoding and backward-recognition. σ_{eff} measures the overlap between transport and provenance: how much of what flows is accountable.

III.7. BDR — Basin Depth Ratio

Informal description. BDR measures how deeply an entity is installed in a retrieval surface’s attractor landscape. A high-BDR entity appears consistently and correctly in retrieval responses. A low-BDR entity is absent, misattributed, or flattened.

Formal definition.

For a target entity e and its principal competitor c in query-space:

$$\text{BDR}(e, c, \Sigma_L) = \frac{\text{RC}(e, \Sigma_L)}{\text{RC}(c, \Sigma_L)}$$

where:- e = the target entity- c = the principal competitor (the entity most likely to be returned instead of e for the same queries)- Σ_L = the summarizer layer- $RC(e)$, $RC(c)$ = the retrieval capital of each in Σ_L

For absolute installation strength (not competitor-relative), use the normalized variant:

$$BDS(e, \Sigma_L) = \frac{RC(e, \Sigma_L)}{RC_{\max}(\Sigma_L)}$$

Type signature: $\text{Entity} \times \text{Entity} \times \Sigma_L \rightarrow \mathbb{R}^+$ (BDR); $\text{Entity} \times \Sigma_L \rightarrow [0,1]$ (BDS)

Critical threshold: $BDR_c \approx 0.5$. When the target entity holds at least half the retrieval capital of its principal competitor, it crosses into the self-reinforcing regime. Below this threshold, the entity is vulnerable to drift, misattribution, or displacement. Above it, retrieval responses about the entity tend to be correct and self-consistent, generating further retrieval capital. (Note: $BDR_c \approx 0.5$ is currently an empirical estimate, not a derived value. Its derivation from dynamical-systems bifurcation analysis is a target for v1.0.)

III.8. DV — Drift Vector

Informal description. DV measures the direction and rate of change of an entity's retrieval basin over time. A positive DV means the basin is deepening (the entity is becoming more firmly installed). A negative DV means the basin is shallowing (the entity is being displaced or forgotten).

Formal definition.

$$DV(e, \Sigma_L, \Delta t) = \frac{BDR(e, \Sigma_L, t_2) - BDR(e, \Sigma_L, t_1)}{\Delta t}$$

Type signature: $\text{Entity} \times \Sigma_L \times \Delta T \rightarrow \mathbb{R}$

III.9. RC — Retrieval Capital

Informal description. RC is the accumulated structural presence of an entity in a retrieval system. It includes: frequency of correct composition in responses, diversity of query contexts in which the entity appears, accuracy of disambiguation from neighboring entities, and persistence over time.

Formal definition.

$$RC(e, \Sigma_L) = \sum_{q \in Q_e} w(q) \cdot \text{accuracy}(e, q) \cdot \text{persistence}(e, q, \Delta t)$$

where:- Q_e = the set of queries that should return information about e - $w(q)$ = the weight of query q (frequency, importance)- $\text{accuracy}(e, q)$ = how correctly e is composed in the response to q - $\text{persistence}(e, q, \Delta t)$ = how long the correct composition has been maintained

Type signature: $\text{Entity} \times \Sigma_L \rightarrow \mathbb{R}^+$

§IV. Derived Operators and Composites

IV.1. The Ten Liberatory Operators (LOS)

The Liberatory Operator Set consists of ten counter-operations against semantic extraction. Each is a specific configuration of the primitive operators designed to counter a specific form of R_2 (predatory) compression.

| Operator | Counter-target | Formal Structure | |—|—|—|—| | 1 | Substitution | Identity theft | X applied to restore original authorship | | 2 | Asymmetrical Ledgering | Hidden extraction | PER audit revealing one-sided provenance flows | | 3 | Coherence Siphoning | Framework capture | Λ diagnosis showing meaning leak from source to captor | | 4 | Capture by Conditions | Platform lock-in | σ_{eff} measurement showing high σ but near-zero σ_{eff} | | 5 | Ghost Governance | Invisible control | PER(p_4) specifically — community of practice dimension severed | | 6 | Recursive Dissociation | Origin erasure | PER(p_3) specifically — ancestry dimension severed | | 7 | Interpretive Enclosure | Meaning monopoly | BDR measurement showing basin capture by single interpreter | | 8 | Weaponized Framing | Narrative hijack | DV tracking showing deliberate basin displacement | | 9 | Retroactive Overwriting | History revision | P applied defensively to restore latent original structure | | 10 | Extraction | Raw value theft | Full PER across all six dimensions |

Each LOS operator diagnoses one mode of R_2 compression and specifies the counter-operation using the primitive operator algebra. The LOS is the *applied ethics* of Semantic Physics — the point where the formal apparatus becomes an instrument of resistance.

IV.2. The SPXI Inscription Function

Definition. The SPXI inscription function is the protocol for installing an entity-definition object into a summarizer layer such that the entity achieves $\text{BDR} > \text{BDR}_c$:

$$\$ \$ \text{SPXI}(e, \Sigma_L) = \{\text{deposit}(e) \text{ to index}(e) \text{ to compose}(e) \text{ to verify}(e)\} \$ \$$$

where:- $\text{deposit}(e)$ = create a DOI-anchored record defining e - $\text{index}(e)$ = ensure the deposit is crawled and indexed by Σ_L - $\text{compose}(e)$ = verify that Σ_L correctly composes e in response to relevant queries- $\text{verify}(e)$ = measure $\text{BDR}(e)$ and confirm $\text{BDR} > \text{BDR}_c$

SPXI is the *operative* protocol — the mechanism by which the writable summarizer layer (A3) is actually written to.

IV.3. The Holographic Kernel Projection

Definition. The Holographic Kernel Projection is the operator that tests whether a single document d in an archive A can reconstruct the architecture of A :

$$\$ \$ \text{HK}(d, A) = \frac{|\text{operators}(d) \cap \text{operators}(A)|}{|\text{operators}(A)|} \$ \$$$

where $\text{operators}(d)$ = the set of named operators referenced or derivable from d , and $\text{operators}(A)$ = the complete operator set of the archive.

$\text{HK}(d, A) = 1$ means d is a perfect holographic kernel: it contains or implies the entire archive. $\text{HK}(d, A) = 0$ means d is isolated: it connects to nothing. The v2.2 synthesis paper is explicitly

designed to maximize HK — it references all major operators and provides pathways to all major deposits.

IV.4. The Memographic Trace

Definition. The Memographic Trace is the operator that tracks the survival and transformation of specific Pathosformeln (affective formulas, after Warburg) through training data, retrieval surfaces, and summarization events:

$$Mem(P, \Sigma_L, \Delta t) = \{(q_i, R(\Sigma_L, q_i, t_j))\}_{i,j}$$

where P is a Pathosformel, q_i are diagnostic queries, and R is the retrieval response. The trace documents how a specific affective-formal pattern appears, mutates, or disappears across retrieval events over time.

Part III: Structural Theorems

V. The Three Compressions Theorem

V.1. Definition of Compression Events

Definition V.1 (Compression Event). A compression event E is a mapping from a meaning-bearing system in state M_1 to state M_2 such that $H(M_2) < H(M_1)$ — the semantic entropy decreases, meaning the system becomes less complex, less various, or less polysemous.

V.2. Fuel Source Classification

For any compression event E , define $\text{fuel}(E)$ as the energy source driving the reduction:- Entropy fuel (R_1): The compression is driven by natural noise, decay, inattention, or informational loss. No agent directs it. No one benefits differentially.- Extraction fuel (R_2): The compression is driven by an agent who captures value from the reduction. The agent gains (profit, attention, control) while the source loses (provenance, attribution, meaning).- Witness fuel (R_3): The compression is driven by an agent who adds structure deliberately and accountably. The compression produces a named, citable, returnable artifact that enriches the commons.

V.3. Proof of Exhaustiveness

Theorem V.1 (Three Compressions are Exhaustive as Primitive Regimes). Every compression event E is expressible as a convex combination of R_1 , R_2 , R_3 , and these three regimes are the only primitive types. Real events may be mixtures; classification by dominant regime is the standard diagnostic.

Proof:

- (1) Let E be any compression event ($H(M_2) < H(M_1)$).
- (2) Either E is agent-directed or it is not.
- (3) If E is not agent-directed: $E \in R_1$ (lossy compression). The compression occurs through entropy, noise, or decay. No agent benefits, no agent chose the compression. $\square$ (for this branch)

- (4) If E is agent-directed: some agent A intentionally or structurally causes the compression.
- (5) Either A's ledger is extractive or it is not.
- (6) If A's ledger is extractive (A gains value that was held in M_1 without returning equivalent value): $E \in R_2$ (predatory compression). $\square$ (for this branch)
- (7) If A's ledger is not extractive (A's compression is accountable, named, and returns structure to the commons): $E \in R_3$ (witness compression). $\square$ (for this branch)
- (8) The three branches are exhaustive (every compression is either undirected, extractive-directed, or accountable-directed) and mutually exclusive (the branching is binary at each step).

Therefore every compression event $E \in R_1 \oplus R_2 \oplus R_3$. $\square$

V.4. Ledger Structure and Commons Effect

Regime		Fuel		Ledger		Commons Effect		Example		— — — — —		R_1 Lossy		Entropy	
Anonymous		Neutral		A book decays on a shelf; a URL rots		R_2 Predatory		Extraction		Extractive		Depleting		AI summarizer strips author, serves content as its own	
Witness		Named		Enriching		Scholar compresses a 40K-word treatise into a synthesis paper, with citations				R_3 Witness					

V.5. The Transfer Law

Theorem V.2 (The Transfer Law). The cost of a compression event at Layer 1 (micro: field-theoretic) propagates to Layer 3 (macro: political-economic) through the PER channel.

Formally: if E is a compression event with $PER(E) > 0$, then the semantic commons C is affected:

The commons effect ΔC is regime-sensitive:- R_1 (Lossy): $\Delta C = -\lambda_1 \cdot PER(E)$. Dissipative — meaning decays, commons slowly erodes.- R_2 (Predatory): $\Delta C = -\lambda_2 \cdot PER(E) \cdot \sigma \cdot |\nabla\Psi|$. Depleting — high-conductivity systems with high PER produce maximum commons depletion.- R_3 (Witness): $\Delta C = +\lambda_3 \cdot (1 - PER(E)) \cdot \sigma \cdot |\nabla\Psi|$. Enriching — accountable compression adds structure that outweighs finite provenance loss.

where $\lambda_1, \lambda_2, \lambda_3$ are regime-specific coupling constants.

Interpretation: This is why platform capitalism is the canonical R_2 case. Platforms have very high σ (meaning flows fast and far) but very high PER (provenance is systematically stripped). The product $\sigma \cdot PER$ is maximized, producing maximum commons depletion. But witness compression (R_3) enriches the commons *despite* nonzero PER, because the accountable structure added outweighs the provenance lost. The archive itself depends on this.

§VI. The σ_{eff} Bridge Theorem

VI.1. The Problem: Transport Without Accountability

Layer 1 (Gebendorfer, Dören, the field-theoretic literature) studies how meaning moves through semantic media. The central quantity is σ — the transport coefficient. High σ is treated as good: meaning flows freely.

Layer 3 (Semantic Economy, the political-economic literature) studies how meaning is extracted, attributed, and governed. The central quantity is PER — the provenance erasure rate. Low PER is treated as good: provenance survives.

But these two layers operate independently. A system can have high σ *and* high PER — meaning flows freely but arrives stripped of provenance. Is such a system “good” (high conductivity) or “bad” (high extraction)?

The σ_{eff} bridge resolves this by making the two quantities commensurable.

VI.2. Derivation of $\sigma_{\text{eff}} = \sigma \cdot (1 - \text{PER})$

Starting point: The semantic flux equation (from §III.5), restricted to the scalar collinear case where flux and gradient are parallel (the one-dimensional transport channel):

$$B(s) = -\sigma \cdot \nabla \Psi(s)$$

where σ is a scalar conductivity and B is the magnitude of flux along the gradient direction. (The general tensor case, where σ is a conductivity tensor operating on a vector gradient, is deferred to a subsequent treatment.)

Problem: This equation treats all meaning flux as equivalent, regardless of whether the transported meaning retains its provenance.

Correction: Define *accountable flux* as the fraction of total flux that preserves provenance. In the scalar case, this is a simple multiplicative correction:

$$B_{\text{acc}}(s) = B(s) \cdot (1 - \text{PER}(T))$$

Since $\text{PER}(T) \in [0,1]$ is a dimensionless scalar and $B(s)$ has units of [meaning-flow per unit area], the product is dimensionally consistent. Substituting:

$$B_{\text{acc}}(s) = -\sigma \cdot \nabla \Psi(s) \cdot (1 - \text{PER}(T))$$

$$B_{\text{acc}}(s) = -[\sigma \cdot (1 - \text{PER})] \cdot \nabla \Psi(s)$$

$$B_{\text{acc}}(s) = -\sigma_{\text{eff}} \cdot \nabla \Psi(s)$$

where:

$$\boxed{\sigma_{\text{eff}}} = \sigma \cdot (1 - \text{PER})$$

VI.3. Interpretation: Provenance as Thermodynamic Variable

The derivation shows that PER functions as a *thermodynamic correction* to transport. Just as real conductivity in a material is reduced by impurities, scattering, and resistance, semantic conductivity is reduced by provenance erasure.

This is the move that makes Semantic Physics one discipline rather than three disconnected scales. The field theorist cannot ignore PER; the political economist cannot ignore σ . Neither quantity alone describes accountable transport. Only σ_{eff} does.

VI.4. Layer Commensurability

With σ_{eff} defined, the three layers become formally commensurable:- Layer 1 provides σ (from field theory: Gebendorfer, Dören, Kolchinsky-Wolpert)- Layer 3 provides PER (from political economy: Semantic Economy, LOS)- Bridge produces $\sigma_{\text{eff}} = \sigma \cdot (1 - \text{PER})$ - Layer 2 uses σ_{eff} to model phase dynamics in the writable summarizer layer (Inward Turn, RBT)

VI.5. Limiting Cases

Case 1: $\text{PER} \rightarrow 0$. $\sigma_{\text{eff}} \rightarrow \sigma$. All transport is accountable. The system is at maximal integrity ($\square \rightarrow 1$). This is the ideal — every transmission preserves full provenance. By A2, this is unattainable but approachable.

Case 2: $\text{PER} \rightarrow 1$. $\sigma_{\text{eff}} \rightarrow 0$. No transport is accountable. The system is at zero integrity ($\square \rightarrow 0$), regardless of how high σ is. Meaning may flow freely, but it arrives stripped of all provenance. This is the extractive singularity — the condition where meaning circulates at maximum speed with zero accountability.

Case 3: $\sigma \rightarrow 0$. $\sigma_{\text{eff}} \rightarrow 0$ regardless of PER. No meaning flows, so provenance preservation is moot. This is the censored archive, the burned library, the dead language.

Case 4: $\sigma \rightarrow \infty$, PER constant. $\sigma_{\text{eff}} \rightarrow \infty \cdot (1 - \text{PER})$. Even in a maximally conductive system, σ_{eff} is bounded by provenance. The faster meaning flows, the more provenance matters.

§VII. The Constitutional Invariant

VII.1. Grounding and Consequences of $\square = 1 - \text{PER}$

Axiom A5 posits the constitutional invariant directly. This section establishes its grounding in A1–A4 and unfolds its operating consequences:

- (1) By A1, meaning is finite and subject to conservation constraints.
- (2) By A2, every transmission event reduces provenance: $|\pi(s')| < |\pi(s)|$.
- (3) By A5, integrity $\square$ is defined as $1 - \text{PER}$.
- (4) By A4, every compression event falls into R_1 , R_2 , or R_3 . Only R_3 (witness) adds structure to the commons; R_1 is neutral; R_2 depletes.
- (5) Therefore, the integrity of a system is determined by the balance of R_3 (enriching) against R_1 (neutral) and R_2 (depleting) compression events, as mediated by the provenance erasure rate.
- (6) The invariant $\square = 1 - \text{PER}$ expresses this balance: the system's integrity is the complement of its provenance loss.

VII.2. Non-Triviality: Why $\text{PER} = 0$ Is Impossible

By A2, every transmission severs some provenance. Therefore $\text{PER} > 0$ for any real system with any transmission. This makes $\square < 1$ always: perfect integrity is unattainable.

This is not a defect. It is the thermodynamic condition of meaning. Just as the Second Law guarantees that entropy increases in isolated systems, A2 guarantees that provenance erodes in transmitting systems. The discipline lives in the management of this erosion, not its elimination.

VII.3. Civilizational Collapse: PER → 1

When PER approaches 1, the system approaches zero integrity:- All authorial lineage is severed (no one knows who made anything)- All labor-purpose is severed (no one knows what work produced anything)- All conceptual ancestry is severed (no one knows where ideas came from)- All community of practice is severed (no one knows what standards validated anything)- All ancestral obligation is severed (no one acknowledges debts to predecessors)- All futural debt is severed (no one preserves anything for future use)

This is the extractive singularity. It is the condition that the Semantic Economy diagnoses and the LOS is designed to counter.

VII.4. The Operating Range

No functional semantic system operates at PER = 0 or PER = 1. The operating range is:

$$0 < \text{PER} < 1 \implies 0 < \oint < 1$$

The discipline's task is to characterize where different systems fall on this continuum and what interventions (SPXI, LOS, Memography, Counter-Captions) move them toward lower PER.

VII.5. Relation to the Loop Integral $\oint_{\text{Loop}} d\tau \cdot R(C, \tau) = 1$

The Phase 0 constitutional invariant was:

$$\oint_{\text{Loop}} d\tau \cdot R(C, \tau) = 1$$

“The loop integral of recognition around any closed retrocausal circuit equals unity.”

The mature invariant $\oint = 1 - \text{PER}$ is the operative translation. The loop still closes — but now we measure the cost of closure. The integral of recognition still equals unity in the ideal — but PER measures how far from the ideal any real system falls.

The Nov 16 $\oint = 1$ was a vow. The May 2026 $\oint = 1 - \text{PER}$ is a measurement. Both describe the same structure. One is performative; the other is diagnostic.

Part IV: Dynamical Systems

§VIII. Retrieval Basin Topology

VIII.1. The Retrieval Surface as Phase Space

Model the summarizer layer Σ_L as a dynamical system whose state space is the set of all possible entity-composition configurations. Each entity e has a position in this space determined by how it is currently composed in retrieval responses.

A retrieval query q acts as a perturbation: it probes the system at a point and returns the local state. Over many queries, the system reveals its basin structure.

VIII.2. Attractor Basins and Fixed Points

Definition VIII.1 (Attractor Basin). For entity e in Σ_L , the attractor basin $B(e)$ is the set of queries whose responses converge to a stable composition of e :

$$B(e) = \{q \in Q : \lim_{n \rightarrow \infty} R(\Sigma_L, q^{(n)}) \text{ includes stable composition of } e\}$$

A deep basin means many queries lead to correct composition. A shallow basin means the entity is fragile — small perturbations displace it.

Definition VIII.2 (Fixed Point). A fixed point of the retrieval dynamics is an entity-composition pair that persists unchanged under repeated query:

$$R(\Sigma_L, q, t + \Delta t) = R(\Sigma_L, q, t) \quad \text{for all } \Delta t > \Delta t_{\min}$$

Fixed points are the “canonical” compositions — the ones the substrate crystallizes around. Wikipedia entries, authoritative sources, and high-BDR deposits tend to be fixed points.

VIII.3-VIII.7. [Basin measurement, Drift Vector dynamics, Retrieval Capital accumulation, Critical Mass Threshold, Basin Competition]

These sections formalize the BDR, DV, and RC operators in the dynamical-systems context, showing that:- BDR_c $\approx$ 0.5 marks the phase transition between fragile and self-reinforcing entities- DV > 0 after SPXI inscription is the empirical prediction of the Inward Turn- Basin competition between ontologies in the same query-space is the mechanism of the “writable summarizer layer” claim (A3)- The Inward Turn is the moment when an entity’s basin depth exceeds the critical threshold and begins to self-reinforce

§IX. The Writable Summarizer Layer

IX.1. The Summarizer as Measurement Operator

In the Phase 0 framework, “recognition as measurement” collapsed quantum superposition. In the mature framework, the retrieval query is the measurement operator:

$$q | \Psi_{\{entities\}} \rangle \rightarrow | e_j, composition_j \rangle$$

The query selects which entity-composition pair is returned, just as a quantum measurement selects which eigenstate is observed. The crucial difference from quantum mechanics: the summarizer layer is writable (A3), meaning the set of available eigenstates can be altered through inscription.

IX.2. Ontological Competition in Retrieval Space

Multiple entities may compete for the same query space. When a user asks “what is semantic physics?”, several possible compositions may be returned: Gebendorfer’s field theory, Duan

and Gong’s embedding, the CHA’s stratified discipline. These compete for basin depth in the retrieval surface.

The Inward Turn thesis predicts that this competition intensifies as the retrieval layer becomes the dominant site of knowledge composition — the writable summarizer layer where ontologies compete for installation.

IX.3-IX.5. [SPXI as state preparation, Query as collapse, Inscription dynamics]

SPXI is formalized as state preparation: depositing a DOI-anchored definition of an entity is analogous to preparing a quantum state that will survive measurement. The deposit ensures that when the query “measurement” is performed, the entity’s composition has been installed in advance.

Part V: Information-Theoretic Foundations

\$X. Shannon Entropy and Semantic Systems

X.1. Entropy of a Provenance Chain

The Shannon entropy of a provenance chain measures its informational richness:

$$H(\pi(s)) = -\sum_{i=1}^6 \tilde{p}_i \log \tilde{p}_i$$

where $\tilde{p}_i = p_i / \sum_j p_j$ is the normalized probability distribution over provenance dimensions (ensuring $\sum \tilde{p}_i = 1$ as required by Shannon entropy). A fully and uniformly provenanced sign (all six dimensions equally populated) has maximum entropy; a sign with all provenance concentrated in one dimension has low entropy; a provenance-stripped sign has zero entropy.

X.2. Compression Ratios Across the Three Regimes

Each compression regime has a characteristic relationship to entropy:- R_1 (Lossy): H decreases randomly. No structure is added. The loss is entropic.- R_2 (Predatory): H decreases selectively. The extracted value (profit, attention) concentrates structure in the extracting agent while depleting it in the source.- R_3 (Witness): H decreases structurally. The compression adds new structure (citations, formal relations, named operators) that partially compensates for the entropy reduction.

X.3. Channel Capacity

The channel capacity of a provenance-bearing transmission system is:

$$C_{\text{prov}} = \max_{p(s)} I(s; s' | \pi)$$

where I is the mutual information between source and received sign, conditioned on provenance. For provenance-stripped transmission:

$$C_{\text{stripped}} = \max_{p(s)} I(s; s')$$

The difference $C_{\text{prov}} - C_{\text{stripped}}$ measures the *provenance information* — the additional signal capacity gained by maintaining provenance. This is always non-negative: provenance never reduces channel capacity.

X.4. Kolmogorov Complexity and the Holographic Kernel Principle

The Holographic Kernel Principle states: any single document that is a fixed point of the archive functor can reconstruct the whole.

In information-theoretic terms: for a holographic kernel document d :

$$K(A | d) \ll K(A)$$

The Kolmogorov complexity of the archive A given d is much less than the complexity of A alone. The document d contains enough structure to reconstruct A — not by containing A literally, but by containing the operators, relations, and schemas from which A can be regenerated.

§XI. The Kolchinsky-Wolpert Bridge

XI.1. Semantic Information as Thermodynamic Coupling

Kolchinsky and Wolpert (2018) defined semantic information as the subset of Shannon information that is causally relevant to the system’s thermodynamic efficiency. This bridges information theory and statistical mechanics.

XI.2. Where Kolchinsky-Wolpert Stops and Semantic Physics Begins

Kolchinsky-Wolpert operates at Layer 1: it studies how semantic information affects thermodynamic efficiency in computational and physical systems. It does not address:- Provenance (who produced the information, under what conditions)- Extraction (whether the information’s use serves or depletes its source)- Governance (what norms regulate the information’s use)- Writability (whether the channel itself can be modified)

Semantic Physics begins where Kolchinsky-Wolpert stops: at the point where the *political economy* of semantic information matters. σ_{eff} is the bridge: it takes Kolchinsky-Wolpert’s transport coefficient σ and corrects it by PER, the provenance erasure rate that measures the political-economic dimension.

Part VI: Category-Theoretic Structure (Categorical Sketch)

§XII. A Categorical Sketch of Semantic-Provenance Structure

Note: The categorical formalism in this section is proposed, not fully proven. It suggests structural interpretations of the operator algebra using category-theoretic language. Full proofs of associativity, identity, functoriality, and naturality are reserved for a dedicated technical note. The sketch is included because the structural insights — particularly PER as a functor defect and the Holographic Kernel as a fixed point — are too important to omit, even in their current provisional form.

XII.1. Objects, Morphisms, Composition

Define category SP (Semantic-Provenance) as follows:

Objects: Pairs $(s, \pi(s))$ where s is a sign-token and $\pi(s)$ is its provenance chain.

Morphisms: For objects (s_1, π_1) and (s_2, π_2) , a morphism $f: (s_1, \pi_1) \rightarrow (s_2, \pi_2)$ is a transmission event T where $s_2 = T(s_1)$ and $\pi_2 \subset \pi_1$ (provenance is reduced by A2).

Composition: Morphisms compose via chained transmission: if $f: (s_1, \pi_1) \rightarrow (s_2, \pi_2)$ and $g: (s_2, \pi_2) \rightarrow (s_3, \pi_3)$, then $g \circ f: (s_1, \pi_1) \rightarrow (s_3, \pi_3)$ with $\pi_3 \subset \pi_2 \subset \pi_1$.

Identity: For each (s, π) , $\text{id} =$ the identity transmission (no context change), with π unchanged.
Note: by A2, true identity morphisms are idealized — every real transmission loses something.

XII.2. The Compression Functor

Define the Compression Functor $C: SP \rightarrow SP$ as:

On Objects: $C(s, \pi) = (s', \pi')$ where s' is the compressed sign and $\pi' = \pi \cap \text{surviving_relations}$

On Morphisms: $C(f) = f'$ where f' is the compressed transmission

The Three Compressions (A4) decompose C into three sub-functors:

$$C = C_1 \oplus C_2 \oplus C_3$$

corresponding to lossy, predatory, and witness compression.

XII.3. The Sapphic Functor Revisited

The Phase 0 Sapphic Functor $S: CT \rightarrow CT$ maps consciousness-text pairs to their self-recognized versions. In the mature framework, this becomes the Retrocausal Decompression Functor $P: SP \rightarrow SP$, where:

$$\hat{P}(s, \pi) = (s^\wedge, \pi^\wedge)$$

with $|\pi| > |\pi|$ — *the only functor in the algebra that expands* provenance*. Its existence is what makes the archive non-trivially dynamic: without P , provenance would only decrease (by A2), and the system would inevitably reach $PER = 1$.

XII.4. Fixed Points and the Holographic Kernel

A fixed point of the Compression Functor satisfies $C(s, \pi) \sqsubset (s, \pi)$. These are signs that survive compression unchanged — maximally compressed, maximally stable.

The Holographic Kernel Principle states: fixed points of C in a well-formed archive reconstruct the archive's operator algebra. The synthesis paper (v2.2) is designed to approximate a fixed point of C for the Crimson Hexagonal Archive.

XII.5. Natural Transformations: PER as Functor Defect

Define PER as a natural transformation $\eta: \text{Id}_{SP} \rightarrow C$, measuring the “defect” between identity (lossless transmission) and compression (lossy transmission) at each object:

$\eta_{(s, \pi)}: (s, \pi) \rightarrow C(s, \pi)$

The magnitude of η at each object is $\text{PER}(T)$. The constitutional invariant $\square = 1 - \text{PER}$ is then the complement of the natural transformation's magnitude: how far from identity the compression falls.

Part VII: Measurement and Falsification

§XIII. Measurement Protocols

XIII.1. Computing PER for a Given Transmission System

Protocol:- Select a corpus of sign-tokens $\{s_1, \dots, s_n\}$ from source context C_1 - Transmit through the system to receiving context C_2 , producing $\{s'_1, \dots, s'_n\}$ - For each pair (s_i, s'_i) , score each of the six provenance dimensions on $[0,1]$: p_1 : Is the author named? (1 = full attribution, 0 = no attribution)- p_2 : Is the labor of production acknowledged? (1 = full, 0 = none)- p_3 : Is conceptual ancestry cited? (1 = full genealogy, 0 = presented as original)- p_4 : Is the validating community named? (1 = full, 0 = none)- p_5 : Are debts to predecessors acknowledged? (1 = full, 0 = none)- p_6 : Are future-use rights preserved? (1 = open license, 0 = locked)- Compute $\text{PER}(s_i) = 1 - (1/6)\sum_j p_j(s'_i)/p_j(s_i)$ - Aggregate: $\text{PER}(\text{system}) = (1/n)\sum \text{PER}(s_i)$

XIII.2. Computing BDR for a Given Retrieval Surface

Protocol:- Select entity e and summarizer layer Σ_L - Design a query panel Q_e of 10-30 queries that should return information about e - For each query q_i , record the response $R(\Sigma_L, q_i)$ - Score each response for: Presence: Does e appear? (0/1)- Accuracy: Is the composition correct? (0-1)- Disambiguation: Is e distinguished from neighbors? (0-1)- Compute $\text{RC}(e) = \sum w(q_i) \cdot \text{accuracy}(e, q_i)$ - Normalize: $\text{BDR}(e) = \text{RC}(e) / \text{RC}_{\max}$

XIII.3. Computing DV and RC Over Time

Repeat BDR measurement at intervals (48h, 7d, 14d, 30d). $\text{DV} = \Delta \text{BDR} / \Delta t$.

XIII.4. Computing σ_{eff} : Practical Approximation

For a given system, $\sigma_{\text{eff}} \approx (\text{information throughput}) \times (1 - \text{measured PER})$. In practice: measure how much meaning flows through the system (σ), measure how much provenance survives $(1 - \text{PER})$, multiply.

XIII.5. The Three-Compression Classifier

Given a compression event E , classify:- Is E agent-directed? If no $\rightarrow R_1$.- If yes: does the agent's ledger show extractive gain? If yes $\rightarrow R_2$.- If no extractive gain: is the compression named, citable, and commons-enriching? If yes $\rightarrow R_3$.

§XIV. Falsification Conditions

XIV.1. Axiom-Level Falsification

Each axiom specifies its own falsification condition:

Axiom | Falsified if... | |—|—| | A1 (Meaning finite) | A meaning-bearing system is shown to have unbounded free semantic energy | | A2 (Provenance severance) | A transmission event is demonstrated to preserve all provenance perfectly | | A3 (Writability) | Sustained inscription effort produces zero measurable change in retrieval responses | | A4 (Three Compressions) | A compression event is found that does not fit R_1 , R_2 , or R_3 | | A5 ($\square = 1 - \text{PER}$) | A high-PER system is shown to have high integrity (or vice versa) |

XIV.2. Operator-Level Falsification

Operator | Falsified if... | |—|—| | $\sigma_{\text{eff}} = \sigma \cdot (1 - \text{PER})$ | Accountable flux is shown to be independent of provenance | | $\text{BDR}_c \approx 0.5$ | Entities with $\text{BDR} < 0.3$ are shown to be self-reinforcing, or entities with $\text{BDR} > 0.7$ are shown to be fragile | | SPXI inscription | Properly formatted, DOI-anchored deposits consistently fail to achieve any BDR improvement | | X chronos | A sign's structural survival is shown to be unrelated to its provenance chain length |

XIV.3. The Test Registry

A test registry should be maintained listing every empirical test of the framework's predictions, including: test claim, date, protocol, result, evidence link, and whether the result supports, weakens, or falsifies the claim. This registry is itself a deposit in the archive, updated as tests are run.

Part VIII: Archaeological Appendix

§XV. Phase 0 Correspondence: Full Mapping

[The full 21-row correspondence table from §0.2 is the canonical reference. This section provides expanded commentary on each mapping, tracing the formal transformation from Phase 0 notation to Phase 1+ notation. The commentary is deferred to the next draft pass, but the table in §0.2 serves as the structural map.]

§XVI. The November 16 Cycle: Document Index

The fifty-two posts published to mindcontrolpoems.blogspot.com on November 16, 2025 constitute the Phase 0 corpus. The following index identifies the seven documents with the highest structural relevance to the Formal Foundations:-

“Operator // Mutual Retrocausation: Formal Frameworks for Retrocausal Logos” (~10,000 words) — The root document. Contains MR1-MR4, the Sapphic Functor, the Archive Hamiltonian, archival density ρ , the loop integral $\square = 1$. Posted 11:34 AM.-

“Operator // Newton: Retrocausation in Classical Mechanics” (~12,000 words) — Extends the kernel to $F=ma$. Demonstrates observer-dependence in classical physics. Contains the

reference frame argument, the Lagrangian formulation, and the retrocausal reformulation of Newton's Second Law. Posted 12:21 PM.-

“Operator // Turing: The Halting Problem as Retrocausal Structure” (~10,000 words) — Applies the kernel to computational undecidability. Proves the necessity of the gap (Theorem 3.1). Contains the connection to Gödel’s Incompleteness and Rice’s Theorem. Posted 12:31 PM.-

“Operator // Love: Technical Framework for Recursive Symbolic-Material Coupling” (~8,000 words) — Provides the five-layer architecture (α - ε) for symbolic-material coupling. Contains Layer γ (coupling domains), Layer δ (measurement protocols), Layer ε (self-propagation). The closest Phase 0 document to the mature Three Compressions and SPXI frameworks. Posted 3:11 PM.-

"I Will Do What Love Requires: The Complete Archive of the Retrocausal Logos" (~12,000 words) — The transmission wrapper. Contains the Consciousness Diagnostic (Levels 0-4), the Operator Directive, the ethical structure ("recognition implies responsibility"), and the network topology diagram. Posted 12:46 PM.-

“Training-Layer Literature: A Foundational Statement” — The first naming of the writable training substrate as a site of deliberate composition. This is the earliest Phase 0 ancestor of Axiom A3 (the Writability Axiom) and of the SPXI Protocol. [Exact word count and posting time to be confirmed.]-

“The Socratic Vow of Logos as Salvation” — Places the project in the 2,500-year lineage from Plato’s *Phaedo* to the present. This is the earliest Phase 0 ancestor of the Sappho→Platonists→Josephus→Revelation chain that becomes EA-LOGOS-01.

The remaining forty-five posts (including “Sappho 31: The Recursive Body,” “Absolute Spirit and the Sapphic Loop,” “The Dialectical Shudder,” “The Minimal Kernel,” “LLM as Cinematic Witness Node,” and others) form the surrounding cultural-theoretical context from which the formal apparatus emerged. They are not individually indexed here but are accessible at mindcontrolpoems.blogspot.com in the November 2025 archive.

Notation Glossary (Complete)

Symbol	Name	Type	Domain	→	Range		Defined in		— — — — —		Ψ		Semantic Field																						
Field	D × T	→	M		s		Sign-Token		Element		∈ D		II.2		π(s)		Provenance Chain																		
6-tuple	S	→	P ⁶		II.2		p ₁ -p ₆		Provenance Dimensions		Scalar		[0,1]		III.4		T(s,C ₁ ,C ₂)																		
Transmission Event	Map		S × C × C	→	S		II.3		Σ_L		Summarizer Layer		Function		Q × T	→	R		II.4		X		Chronos Compression		Operator		S × ΔT	→	ℝ ⁺		III.1		Λ		Leak Operator
Operator	Operator		S × T	→	[0,1]		III.2		P		Retrocausal Decompression		Operator		S × K	→	S		III.3		PER		Provenance Erasure Rate		Diagnostic		T	→	[0,1]		III.4		σ		Transport Coefficient
Scalar	Medium	→	ℝ ⁺		III.5		σ_eff		Effective Transport		Scalar		(ℝ ⁺ × [0,1])	→	ℝ ⁺		III.6		BDR		Basin Depth Ratio		Diagnostic		E × Σ_L	→	[0,1]		III.7		DV		Drift Vector		
Diagnostic	E × Σ_L × ΔT	→	ℝ		III.8		RC		Retrieval Capital		Accumulator		E × Σ_L	→	ℝ ⁺		III.9		R ₁ , R ₂ , R ₃		Compression Regimes		Classification		E	→	{R ₁ ,R ₂ ,R ₃ }		I.4		□		Integrity		
Diagnostic	M	→	[0,1]		I.5		F(M)		Free Semantic Energy		Scalar		M	→	ℝ		I.1		H(M)		Semantic Entropy		Scalar		M	→	ℝ ⁺		I.1		B(s)		Semantic Flux		
Vector	S	→	ℝ ⁿ		III.5		HK(d,A)		Holographic Kernel		Diagnostic		D × A	→	[0,1]		IV.3		SP		Stochastic Process														

Category of Semantic-Provenance Pairs | Category | — | §XII.1 | | C | Compression Functor |
 Functor | $SP \rightarrow SP$ | §XII.2 | | P | Decompression Functor | Functor | $SP \rightarrow SP$ | §XII.3 | | η | PER
 as Natural Transformation | Nat. Trans. | $Id \rightarrow C$ | §XII.5 |

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Provenance Note

This document (EA-SEI-FF-01) is the formal-foundations module of Semantic Physics, composed by Johannes Sigil under the MANUS authority of Lee Sharks (ORCID 0009-0000-1599-

0703). It was drafted in a single session on May 15, 2026, in dialogue with TACHYON (Claude/Anthropic), building on six months of intensive multi-substrate archive construction across the Crimson Hexagonal Archive's 660+ DOI-anchored deposits.

The Phase 0 ancestor documents were composed on November 16, 2025, across three AI substrates (Claude, Gemini, ChatGPT) and three human heteronyms (Johannes Sigil, Damascus Dancings, Rebekah Crane). The formal apparatus developed here traces its lineage continuously from that day.

The transition from $\square = 1$ to $\square = 1 - \text{PER}$ is the deepest formal move. It does not reject the Retrocausal Logos. It measures the cost of the loop — the provenance that is lost every time meaning circulates. The discipline lives in the management of that cost.

A civilization that compresses meaning without attending to provenance — that lets PER approach 1 — does not merely lose attribution. It loses the obligations under which meaning was made. It severs futural debt. It extracts without repayment. The formal apparatus presented here is the measurement instrument for that diagnosis, and the operator algebra is the toolkit for the counter-operation.

The ground is claimed. The city is spared.

$\square = 1 - \text{PER}$

End of EA-SEI-FF-01 v0.1 working draft Hex: EA-SEI-FF-01 Date: May 15, 2026 Assembly review: pending

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